Scrum Documentation – Red Monopoly (Loan System Implementation)

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Project: Red Monopoly – Option 1: Mortgage & Bank-Loan System
Course: Software Engineering 2 (IS2)
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2.1 Structure and Operation of the Scrum Team

- **Team Composition:** Solo project — only one member: Adam Ankoud.
- **Coordination & Tools:**
  - ChatGPT was used as a planning assistant.
  - Java development was done with IntelliJ IDEA.
  - UML diagrams were designed with PlantUML.
  - Version control with Git and GitHub.
- **Work Style:** Fully remote, asynchronous work with daily personal sprints and weekly review cycles.

Agile Inception Reflection: Originally, the goal was to implement the entire Monopoly rule set, but due to scope, Option 1 was selected to focus the development. This choice allowed a deeper implementation and more refined feature delivery.

2.2 User Stories

ID	*User Story*	*Description*	*Priority*	*Weight*	*Status*
US1	Take a loan	As a player, I want to take a loan with interest to boost my balance	High	5	Done
US2	Repay a loan	As a player, I want to repay my loan if I have enough money	High	5	Done
US3	Interest Over turn	Interest should accumulate	Medium	3	Done

		every turn			
US4	Show Loan Buttons	As a player, I want UI buttons to take/repay loans	Medium	2	Done
US5	Save Loan in GameState	Loans must persist when saving/loading the game	Medium	4	Done

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2.3 Sprint Plannings

- **Sprint 1:** Define Option 1 feature scope. Design Loan class and responsibilities. Estimate work units.
- **Sprint 2:** Implement Loan and BankService logic. Connect to Player.
- **Sprint 3:** Integrate turn logic and GUI. Test foreclosure.
- **Sprint 4:** Finalize GUI, hook into GameManager, balance updates.

Each sprint was roughly 2-3 days long, with nightly retrospectives and planning.

.4 Sprint Reviews

- After Sprint 1: Design made sense and scaled well. Player responsibilities were kept clean.
- After Sprint 2: Refactored Player to better support ownedProperties, totalAssets.
- After Sprint 3: GUI worked but UX was not intuitive; tooltips and button spacing improved.
- After Sprint 4: Feature complete. Full gameplay test passed. No runtime issues found.

Definition of Done:

- Compiles and runs without crash
- Feature is visible and functional in the GUI
- Feature works in both offline and network play
- Code reviewed for structure and readability

2.5 Sprint Retrospectives

- **Sprint 1:** Too much time was spent exploring unused features. Future sprints focused more tightly.
- **Sprint 2:** Clearer goals helped reduce rework. Divided logic cleanly into BankService.
- **Sprint 3:** GUI took more time than expected. ChatGPT helped generate boilerplate quickly.
- **Sprint 4:** Final test loop helped catch missing updates to balance labels.

2.6 Product Backlog Evolution

Originally, the game had a backlog for multiplayer networking, AI improvements, and real-estate mechanics. This backlog was trimmed to focus solely on Option 1, with loan mechanics being the central theme.

Removed Backlog Items:

- Online trade system
- Mortgage auction feature (separate from foreclosure)
- Richer AI decision trees

Added/Final Backlog:

- Loan class
- BankService
- GUI button integration
- Interest logic
- Foreclosure
- Save/load loan state

2.7 Sprint Backlogs

Sprint	Task	Status
Sprint 1	Design Loan class, modify Player	Done
Sprint 2	Implement BankService logic	Done
Sprint 3	Hook into GameManager, test per-turn logic	Done

Sprint 4	Add UI buttons and dialogs	Done
Sprint 5	Final test, polish UX, write documentation	Done

2.8 Individual Contribution – Adam Ankoud

- Wrote all source code for the loan system (Loan.java, BankService.java)
- Refactored Player.java to support loan field and methods
- Integrated loan logic into GameManager and GameWindow
- Created UML diagrams with ChatGPT & PlantUML
- Designed GUI logic using Swing
- Handled version control, testing, and final packaging

Personal context: As the sole developer, I managed all phases of the project, with ChatGPT serving as a technical assistant.

2.9 Conclusions and Self-Assessment

This project was a strong exercise in feature-focused agile development. By isolating Option 1, I was able to deliver a complete, polished game feature in a short time, balancing both UI and backend logic. The integration of AI tools like ChatGPT helped accelerate development and structure ideas clearly. I learned to work more iteratively, deliver usable increments fast, and test edge cases more rigorously.

I'm proud of completing this module alone, and I believe the feature I delivered would be a valuable extension to the Red Monopoly Game.

Screenshots:

This is the Loan Button